

Memo

To: President Crawford of Miami University

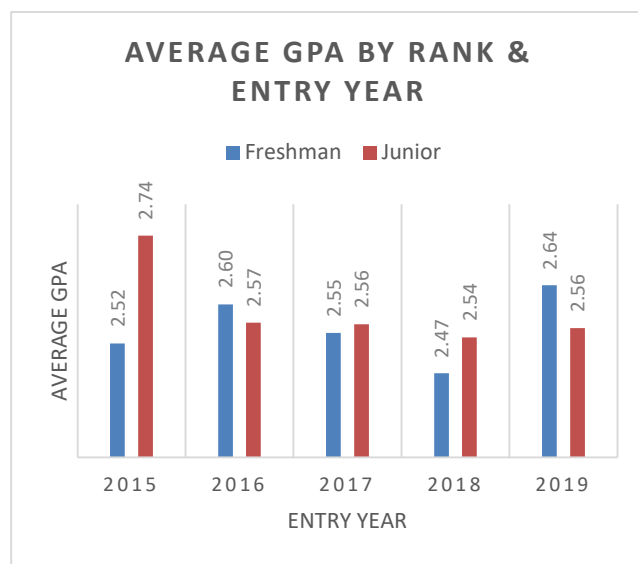
Date: 11/25/2024

Re: University Implemented Programs Analysis

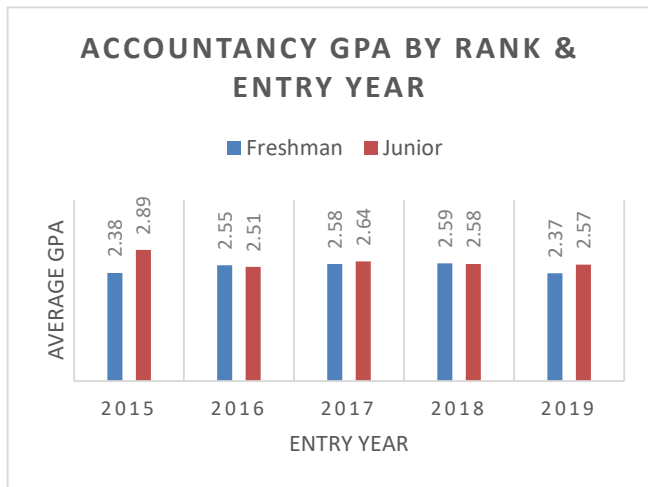
Students at Miami University undergo significant changes during their tenure here. In 2015 the university implemented programs to assist students in hopes that their GPA would improve during their time at the university. This memo provides an analysis of GPA trends among students from their freshman to junior year for entry years from 2015 to 2019. By comparing this data, we will be able to evaluate the effectiveness of these initiatives.

Analysis 1: Average GPA Change by Entry Year

When analyzing GPA for each entry year class, there is an increase in the GPA from freshman year to junior year for 60% of the classes in the years of 2015, 2017, and 2018, while 40% of the classes see a decrease in GPA from freshman year to senior year. 2015, the first year that the programs were implemented, noticed the greatest increase in GPA from freshman to junior year. The most recent entering class observed saw the greatest decrease in GPA from freshman to junior mostly due to external factors. Overall, the majority of the entry years benefited as a result of the implemented programs; however, they were most productive during the first year they were implemented and the most harmful is the most recent year. Furthermore, as there is no clear trend, it is difficult to say whether there is a causation relationship between the programs and improvement of GPA. But further analysis will narrow down the populations being analyzed and determine if there are specific groups that are being affected by the programs.

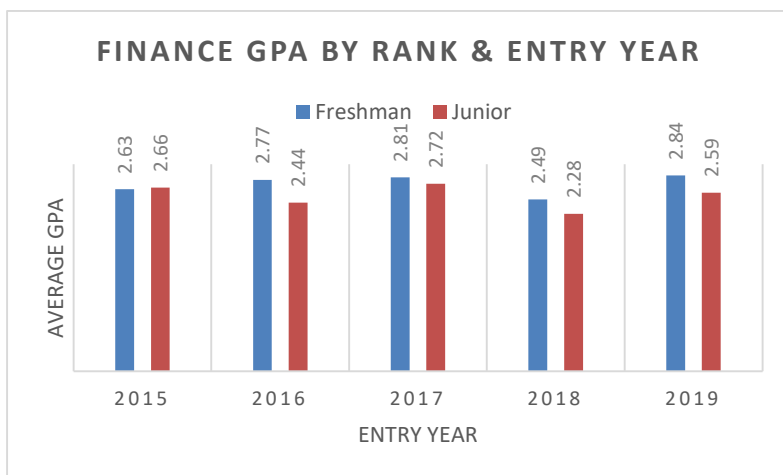
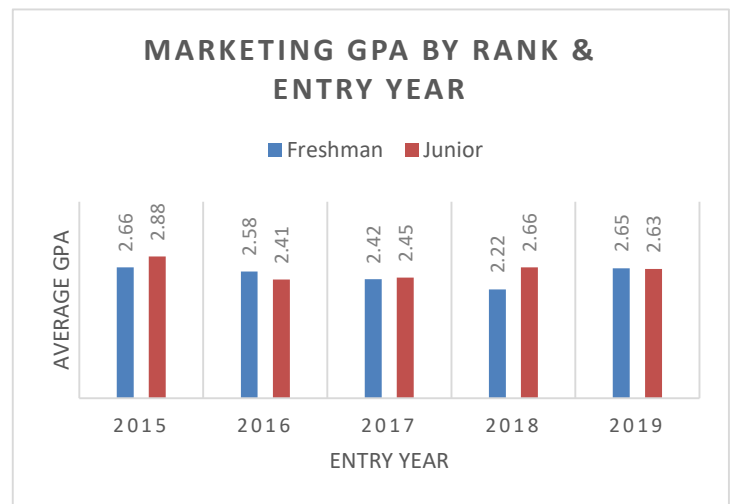


Analysis 2: Average GPA Change by Entry Year per 3 Most Populated Farmer Majors



Accounting majors consistently saw an increase in their GPA since the implementation of the programs in 2015. Although there was only a slight decrease for those who entered Miami in 2016, every other entry year saw improvement from their freshman to junior year.

The impact amongst marketing majors has had varying results, with there being larger effects on the average GPAs compared to the average GPAs of accounting majors. Although both 2015 and 2018 showed sizeable increases in the average GPA, 2016, 2017, and 2019 all showed impactful drops. Overall, there is no consistent trend to conclude if there was a real impact because of the programs. However, marketing majors replicated what the general Miami population with there being a significant increase in 2015 and a considerable decrease in 2019.



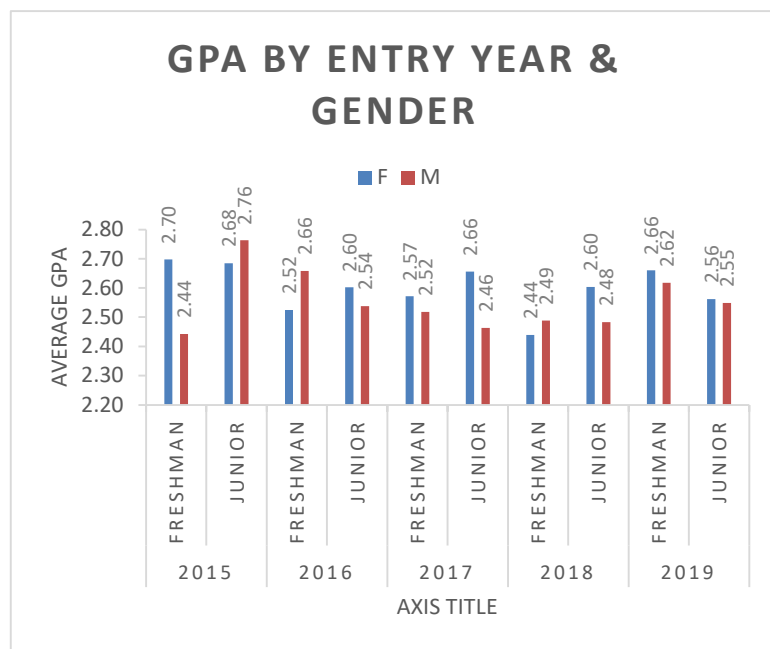
A consistent trend is present for finance majors as for each entry year, the average GPA from freshman to junior year decreased, showing a negative correlation between the effects of the programs and their GPA over their time at Miami University.

Upon these analyses we observed the program had the greatest positive impact amongst those majoring in accountancy while those majoring in finance saw a consistent decrease in GPA between freshman and junior year. The impact amongst the marketing major has had varying results between entering class; thus, we aren't able to make a conclusion whether there was a direct impact of the program on the marketing majors. Furthermore, although there may be correlation for the impact on finance and accounting majors which had opposite reactions, more data would need to be collected to determine if there are outside factors impacting these GPAs. For example, differences in courses and grading styles could play a factor in GPAs over time as the students in those majors take classes more focused on their individual majors after freshman year.

Analysis 3: Average GPA Change by Entry Year and Gender

It is observed that females have been positively impacted from the implementation of these programs. Every entry year, excluding 2019 and 2015 which saw a decrease and little effect respectively, saw an increase in the average GPAs of female students at Miami. However, the programs effects have been less desirable with males. Except for students of entry year 2015 which saw an increase in average GPA for males, every other entry year consistently decreased. Overall, we can say that there is correlation between average GPA change by entry year and gender, with males potentially being hindered by the programs and females seeing the benefits of them.

Moreover, a continuing trend can be seen with gender as entry year of 2015 for both genders saw an increase in their average GPA, while the entry year of 2019 collectively saw a decline in average GPA.



Conclusion:

The implementation of various programs for the entering classes from 2015 to 2019 has had vary affects, based on gender, declared majors and entry year. Based on the data, we cannot conclude that there is correlation between the change in average GPA and the general population of Miami and for marketing majors. However, we can conclude correlation for accounting and finance majors and for genders, with accounting majors and females both being benefitted by the programs, and finance majors and males both being harmed by the initiatives

that were implemented in 2015. The program's effectiveness seems to have lessened each time a new class enters the university. 2015 appears to have been the greatest improvement for GPAs in each observed scenario. However, for students that entered Miami in 2019, the average GPA typically declines.

Due to the variances in impact among the years, further investigation might be needed to conclude whether these programs had a casual impact on the students who entered after 2015. One specific thing that could be analyzed more deeply is the entry year classes and the variations between them. For example, looking at high school GPA's and other factors that signal performance and can make conclusions based on the general populations of those entry years. From there, we would be able to determine if the variations between the entry years are significant enough to impact the results of this. If so, then it would be more difficult to analyze whether the programs did indeed have an impact, and we would have to separate these factors to focus on the effectiveness of the programs. However, if the variation between the entry year classes and their previous performance had a negligible impact of the data that we analyzed, then it would be difficult to say if these programs had an actual impact based on the varied data per entry year. Nonetheless, we were able to see that some trends based on gender and some Farmer School of Business departments were produced and we can see the impact of the programs. But the data on the overall Miami population has too many varying results that it is difficult to determine if the whole population was truly impacted by these programs.